

BECKLEY, OH, USA

WMO: 724295

Lat: 39.840N

Lon: 83.840W

Elev: 320

StdP: 97.54

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 63888

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.3	-21.3	0.6	-16.5	-18.3	0.8	-13.8	12.2	-1.3	11.0	-1.6	3.1	280	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.4	24.2	31.1	23.5	29.9	22.8	26.2	30.2	25.2	29.3	24.3	28.1	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.1	28.8	24.0	19.7	27.4	23.1	18.6	26.5	83.3	29.4	78.6	29.2	74.7	28.2	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.1	8.1	DB	-22.4	34.1	3.9	1.9	-25.2	35.5	-27.4	36.6	-29.6	37.7	-32.4	39.1	(4)
(5)			WB	-22.4	27.3	3.8	1.3	-25.1	28.3	-27.4	29.0	-29.5	29.8	-32.3	30.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-2.8	-1.3	5.1	11.3	17.7	21.9	23.1	22.3	19.1	12.7	5.3	0.7	(6)
(7)		DBStd	10.55	6.85	7.02	6.48	5.30	4.67	3.15	2.75	2.71	4.04	5.28	5.48	5.54	(7)
(8)		HDD10.0	1434	401	322	180	47	2	0	0	0	1	32	159	291	(8)
(9)		HDD18.3	3081	657	551	410	218	72	7	1	3	38	188	391	545	(9)
(10)		CDD10.0	1919	2	4	29	85	240	357	407	382	275	115	19	4	(10)
(11)		CDD18.3	524	0	0	1	6	51	114	149	127	63	13	1	0	(11)
(12)		CDH23.3	4577	0	0	13	67	474	959	1289	1033	611	129	2	0	(12)
(13)		CDH26.7	1448	0	0	2	9	128	322	442	307	207	33	0	0	(13)
(14)	Wind	WSAvg	3.6	4.6	4.5	4.4	4.4	3.7	3.1	2.4	2.1	2.5	3.4	3.9	4.3	(14)
(15)	Precipitation	PrecAvg	1041	80	56	75	98	120	113	122	82	79	73	68	77	(15)
(16)		PrecMax	1335	238	116	175	190	204	197	214	162	172	133	115	197	(16)
(17)		PrecMin	752	32	19	24	28	35	55	42	16	21	18	17	25	(17)
(18)		PrecStd	155	45	26	34	38	50	35	41	35	43	37	27	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	19.2	25.2	27.9	31.3	33.4	34.5	33.1	33.3	30.2	22.3	17.8	(19)
(20)			MCWB	14.4	14.9	17.7	17.3	21.7	23.5	27.0	25.0	22.9	22.0	14.7	16.0	(20)
(21)		2%	DB	13.4	15.1	21.5	25.5	29.6	31.7	32.2	31.1	31.1	26.8	19.3	14.7	(21)
(22)			MCWB	11.8	12.6	15.8	16.8	20.5	23.2	25.2	24.3	22.8	19.2	13.8	12.9	(22)
(23)		5%	DB	10.2	12.7	17.9	22.9	28.3	30.2	30.7	29.7	29.2	24.1	16.5	12.2	(23)
(24)			MCWB	8.8	10.5	13.5	15.2	20.1	22.4	24.2	23.4	22.0	17.8	12.2	11.0	(24)
(25)		10%	DB	7.1	9.0	15.5	20.4	26.5	28.6	29.3	28.4	27.1	21.6	13.9	9.4	(25)
(26)			MCWB	5.2	6.7	12.3	14.5	19.6	21.6	23.7	22.6	20.3	16.8	10.3	7.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.7	18.1	19.5	23.9	26.1	28.1	27.1	26.5	22.9	17.5	16.1	(27)
(28)			MCD B	16.2	18.7	23.1	24.1	29.1	29.9	31.8	30.7	29.8	28.5	21.5	17.4	(28)
(29)		2%	WB	12.1	13.2	16.3	17.8	22.4	24.8	26.7	25.6	24.2	21.1	15.0	13.4	(29)
(30)			MCD B	13.4	14.4	20.1	22.4	27.4	29.6	30.8	29.3	29.1	24.3	17.7	14.3	(30)
(31)		5%	WB	9.1	10.7	14.7	16.7	21.4	24.0	25.6	24.6	23.0	19.3	12.8	11.0	(31)
(32)			MCD B	10.0	12.6	17.1	20.7	26.0	28.5	29.6	28.3	27.0	22.4	16.0	12.2	(32)
(33)		10%	WB	5.2	6.8	12.6	15.4	20.4	23.0	24.6	23.7	21.8	17.6	10.7	7.8	(33)
(34)			MCD B	6.4	8.9	15.2	19.5	24.4	26.6	28.0	27.0	25.2	20.9	13.5	9.3	(34)
(35)	Mean Daily Temperature Range	MDBR		8.5	9.1	10.5	12.0	11.7	11.0	10.7	11.4	12.5	11.7	10.1	7.7	(35)
(36)		5% DB	MCD BR	10.3	12.2	13.8	15.1	13.6	12.9	12.0	12.3	14.7	14.5	13.3	11.1	(36)
(37)			MCWBR	9.4	10.2	9.0	8.4	6.3	6.0	6.0	6.2	7.1	7.8	8.8	10.1	(37)
(38)		5% WB	MCD BR	10.0	11.7	11.6	12.0	11.4	10.9	11.0	10.6	12.2	12.1	11.4	10.8	(38)
(39)			MCWBR	9.7	10.4	8.4	8.2	6.2	5.7	6.5	5.7	6.4	7.2	8.7	10.4	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.300	0.324	0.345	0.388	0.434	0.443	0.446	0.442	0.389	0.354	0.322	0.303	(40)
(41)		taud		2.443	2.347	2.368	2.265	2.174	2.200	2.196	2.190	2.319	2.412	2.460	2.454	(41)
(42)		Ebn at Noon		868	891	909	886	848	839	831	824	850	846	834	836	(42)
(43)		Edh at Noon		80	101	111	131	147	143	143	140	115	94	79	74	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.69	2.56	3.55	4.79	5.50	6.07	5.92	5.46	4.45	3.12	2.10	1.43	(44)
(45)		RadStd		0.15	0.30	0.31	0.39	0.46	0.40	0.34	0.28	0.41	0.28	0.26	0.14	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+63

Nomenclature: See separate page